Range of α-particles & setting up $\alpha - \gamma$ coincidence

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I. OBJECTIVE

1. To set up $\alpha - \gamma$ coincidence measurements using a ^{241}Am source.

II. PROGRESS

A. Week 4

- We familiarized ourselves with the FPGA working principles.
- We set up the high-voltage power supply and NaI scintillator detector for γ-ray detection.
- We verified the signals from both the α and γ detectors using an oscilloscope.
- We avoided using external amplifiers since their processing delays could affect the timing required for coincidence [1].
- We initially could not establish communication between the Red Pitaya and the computer due to two Ethernet interfaces.

B. Week 5

- With the help of Purvesh Garge, a PhD student in the Quantum Optics Lab, we identified the IP-address mismatch as the cause of the communication problem.
- We configured the Ethernet connection by assigning the Red Pitaya and the computer IP addresses in the same subnet using a USB–LAN adapter. This established direct communication between the computer and the Red Pitaya.
- We verified the operation of the Red Pitaya using its built-in oscilloscope.

C. Week 6

- We used the data-acquisition code from the GitHub repository for $\alpha - \gamma$ coincidence measurements [2].
- We installed Vivado 2015.4 to compile the FPGA source files. Since the required Vivado license was unavailable, we used the pre-compiled FPGA bitstream provided in the repository.
- Since the Red Pitaya runs Alpine Linux on an ARM processor, we compiled the `agc.cpp` source on Ubuntu as an ARM-compatible executable and transferred it to the Red Pitaya along with the FPGA bitstream.
- The `agc` executable generates `agc_conf.txt`, where parameters such as the α and γ thresholds and coincidence window can be specified.

D. Week 7

- The α-signal observed on the oscilloscope had an average peak height of approximately 140 mV. The Red Pitaya input range in HV mode is 0–20 V, while the AGC uses 8191 amplitude channels. Thus, the α pulse occupied only a small number of channels, limiting the pulse-height resolution. We therefore switched the α preamplifier to its β setting, which provides approximately 10× higher gain.
- The SD-card filesystem used by the Red Pitaya was mounted as read-only. Hence, files transferred to the temporary root filesystem were lost after a reboot.
- Whenever we start the data acquisition, the Red Pitaya disconnected. We spent two days debugging the DAQ software and testing different stages of its operation, but could not resolve the issue.
- Due to the limited time available, we switched to a Teledyne LeCroy WaveSurfer oscilloscope which offers SMART Trigger. We used an AND gate between the two detector channels with a defined coincidence window to record $\alpha - \gamma$ coincidence events (as shown in Fig 1).

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FIG. 1. Example of a measured $\alpha-\gamma$ coincidence event recorded using the oscilloscope. The blue trace (Channel 3) corresponds to the α-particle detector signal, while the green trace (Channel 4) corresponds to the γ-ray detector signal.

- The coincidence count rate was low, allowing only a small number of events to be acquired. The coincidence window, α and γ thresholds, and source–detector geometry therefore need to be optimized before the final measurements.
- We aim to perform the final optimization and take the measurements during the mid-semester break.

[1] Mario Vretenar, Nataša Erceg, and Marin Karuza. Energy-resolved coincidence counting using an FPGA for nuclear lifetime experiments. *American Journal of Physics*, 87(12):997–1003, 12 2019.

[2] Mario Vretenar, Nataša Erceg, and Marin Karuza. Agc: Alpha-gamma coincidence data acquisition system for Red Pitaya, 2019.